

Digital Twin in Industry: Concepts, Architecture and Applications

Samuel Baraldi Mafra



“

When wireless is perfectly applied, the whole earth will be converted into a huge brain, which in fact it is, all things being particles of a real and rhythmic whole.



Nicola Tesla

What is a Digital Twin?

DT

A digital twin is a digital copy of a physical product, service, or process. This digital twin works as a detailed simulation of objects or operating models, replacing the creation of real prototypes.

Lecture objectives and roadmap

1

Concept and fundamentals

Define digital twin and differentiate it from simulation, digital model and digital shadow

2

Architecture and standards

Understand the 6-step architecture and the ISO 23247 framework

3

Implementation requirements

Identify what is needed to implement a digital twin

4

AI and industrial metaverse

Discuss convergence with generative AI and the industrial metaverse

5

Security and sustainability

Analyze data governance and environmental KPIs

6

Real cases

Analyze cases in Formula 1, manufacturing and marketing

Digital Twin vs. Simulation

Every digital twin uses simulation — but not every simulation is a digital twin.

Simulation

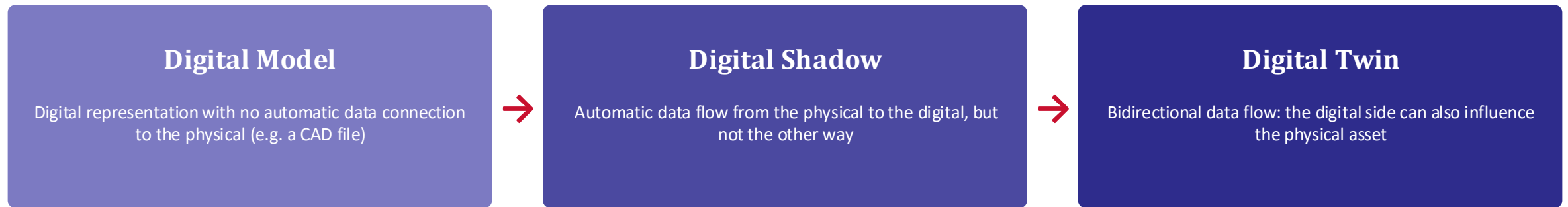
- Generic model of a process or system, with no permanent link to a specific physical asset
- Runs at a point in time, to answer a specific question ("what happens if...")
- Does not receive real-time data from the physical world
- Useful for exploring hypothetical scenarios, even before the asset exists

Digital Twin

- A replica linked to a specific, identifiable physical asset
- Continuously updated with real-time data (bidirectional synchronization)
- Follows the asset throughout its entire life cycle
- Allows not only simulating, but also monitoring and, in some cases, controlling the real asset

Digital Model, Digital Shadow and Digital Twin

Three levels of synchronization between the physical and digital worlds:



Increasing degree of synchronization between the physical and digital worlds

Types of digital twin

1

Component Twin

Represents an individual part or component (e.g. a bearing)

2

Asset Twin

Represents a complete piece of equipment (e.g. a motor, a turbine)

3

System Twin

Represents the interaction between multiple assets (e.g. a production line)

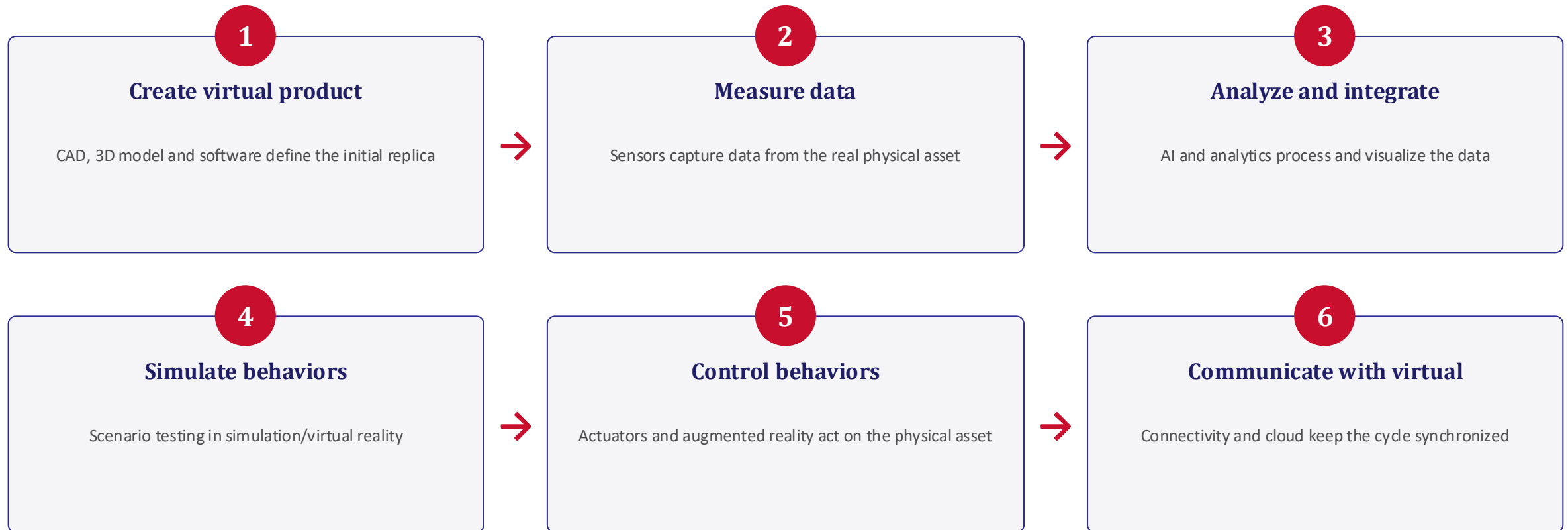
4

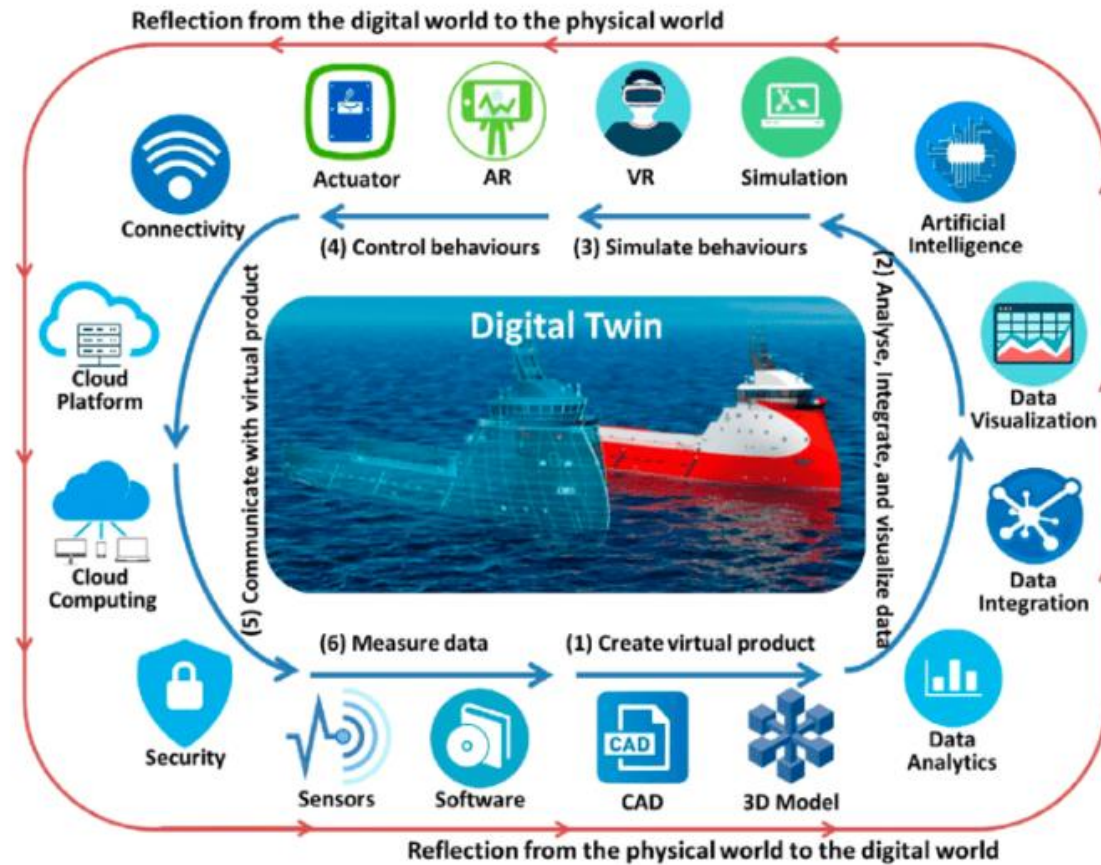
Process Twin

Represents an entire business process (e.g. the whole production chain)

Digital twin architecture (6 steps)

A continuous cycle between the physical product and its virtual replica:





Defining a digital twin — 5 criteria

1

Represents physical assets

It is a digital model representing assets in the physical world

2

Is not just a data model

Must include relational interaction, not just static data

3

Looks and feels like the real thing

Faithfully reflects the physical system's behavior

4

Connects to real-time data

Ensures the model mirrors reality continuously

5

Simulates with varying fidelity

Projects future behaviors with different degrees of precision

ISO 23247 — the international framework

- International standard: "Automation systems and integration — Digital twin framework for manufacturing"
- Defines a common reference architecture for digital twins in manufacturing
- Part 6 (2026) deals specifically with digital twin composition
- Core goal: enable interoperability between digital twins built by different vendors
- A growing reference point in academic research on industrial digital twins

Digital twin composition (ISO 23247-6)

1

Integrated

Digital twins combined into a cohesive implementation, usually from the same vendor

2

Unified

Digital twins from different sources combined under a common data model

3

Federated

Independent digital twins that interoperate while keeping their own architectures

Implementation requirements

1

Define the system

Understand and capture data about the entity to be digitized

2

Identity of things

Each digitized system needs a unique identity to act autonomously

3

Clear objectives

Know how the technology will help achieve specific goals

4

Start small

Monitor a simple component or a single IoT device and expand over time

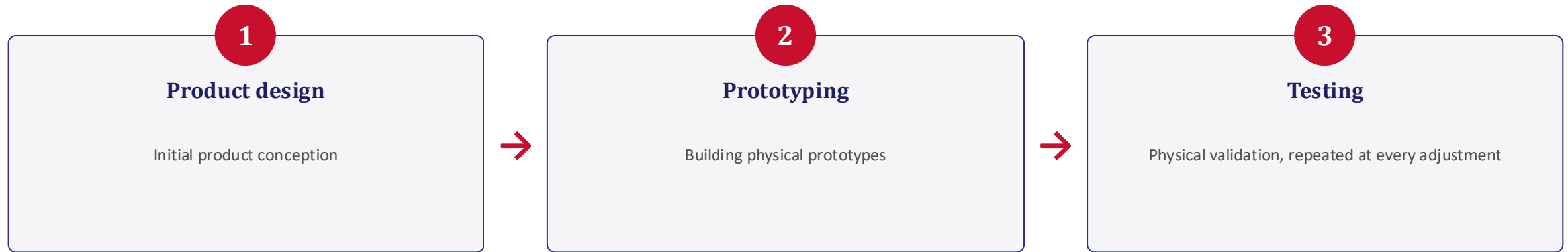
5

Security considerations

The digital twin will constantly transfer sensitive data

Current production model (without a digital twin)

In the current production model, this demands extra time and resources:



With a digital twin, the process is different

Without a digital twin

- Depends on physical prototypes
- Longer test cycles
- Less exploration of design alternatives

With a digital twin

- Simulates real usage conditions before production
- Identifies problems and improvements earlier
- Allows testing more diverse and creative solutions

In the broader commercial world

- Combines high-performance analytics expertise with advanced sensors, simulation and software
- Generates thousands of product variations from that combination
- This makes it possible to deliver the right product to the right market

Interaction between an object and its digital twin

- The advance of IoT allows any product to become smart, capable of collecting usage data and interacting with its virtual copy
- In the future, each object will be unique, with its own digital identification
- Example: an industrial motor, continuously monitored by its digital twin



Data security and governance

- A digital twin depends on a constant flow of data from the physical asset to the digital ecosystem
- This expands the exposure surface of sensitive operational data
- Requires access control, encryption and clear data-retention policies
- Who owns the data: the asset manufacturer, the operator, or the platform developer?
- Data governance must be defined before implementation, not after

Generative AI and "what-if" simulations

- AI applied to digital twins goes beyond monitoring: it suggests and tests scenarios automatically
- "What-if" simulations assess the impact of changes before applying them to the physical asset
- AI models can predict critical quality parameters from hypothetical scenarios
- Brings the digital twin closer to the prescriptive maintenance discussed in the previous lecture

Industrial metaverse

- Elevates the digital twin from an isolated element to an interconnected network of assets
- Covers the entire production ecosystem, not just one specific piece of equipment
- Combines AI, complex simulations and extended reality (XR) in a shared environment
- Adopting standards such as ISO 23247 becomes a condition for interoperability at scale

Digital twin and Industry 5.0

1

Human-centric

Cognitive assistance and more natural interfaces for the operator

2

Sustainable

Using digital twin data to reduce waste and energy consumption

3

Resilient

Ability to simulate and respond quickly to disruptions

Environmental KPIs and energy efficiency

- Digital twins increasingly incorporate environmental KPIs, not just production ones
- Energy and materials inventory data feeds efficiency decisions
- Sustainability becomes a design criterion, aligned with Industry 5.0
- Still a developing research area, with few consolidated standards

Critical discussion: limits of the digital twin

- Model fidelity depends directly on the quality of the input data
- Keeping the model synchronized over the asset's entire life cycle is an underestimated challenge
- Integration with legacy systems is not always trivial
- Who ensures the digital twin still faithfully represents the real asset years later?

Virtual representation and life cycle

- The virtual representation of objects supports life-cycle monitoring of products, machines and equipment
- Every stage — production, use, maintenance, disposal — can be tracked digitally



Communication and visibility via digital twin

- Lets managers see multiple aspects of the supply chain and operations
- Includes inventory, production capacity, average cycle time and product delivery
- This data is fundamental to the decision-making process



Case: Formula 1 — from 1986 to 2020

- F1 cars evolved from simple structures into systems with hundreds of sensors
- Real-time telemetry feeds digital twins used during the race
- Teams simulate pit-stop strategy, tire wear and aerodynamics before deciding in real time
- Race decisions today combine data from the physical car with its digital twin

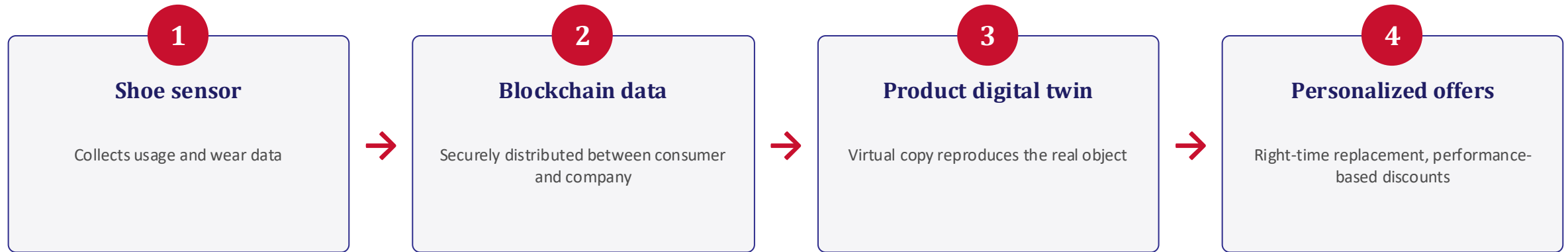


2020



Case: Digital twin applied to marketing

A running shoe equipped with smart sensors:



What these cases have in common

- Real-time data connected to immediate operational or commercial decisions
- Convergence between physical sensing and the digital model
- Focus on anticipating problems or opportunities before they become costly
- Scalability and personalization as core benefits

Benefits: reliability and reduced risk

1

Higher reliability and availability

Monitor, simulate and control an asset as a strategy to improve system performance

2

Reduced risk

Protect health, safety and the environment, reducing incidents and unplanned downtime

Benefits: maintenance cost and production

1

Lower maintenance costs

Predict problems before failures and schedule repairs without affecting production targets

2

Increased production

Real-time view of asset performance to react to customization and minimize supply-chain impact

Trends and open research questions

- Full standardization via ISO 23247 in multi-vendor environments
- Generative AI natively embedded in industrial digital twins
- Digital twins as the foundation of the industrial metaverse at scale
- Formal validation of digital twin fidelity over time
- Digital twin as a central sustainability tool (Industry 5.0)

References, links and videos

- Introduction to Digital Twin: Simple, but detailed — <https://youtu.be/RaOejcczPas>
- Digital Twins in F1 Racing Throw Down Big Gains for Mercedes-AMG Petronas Motorsport — https://youtu.be/WqX5b_HHHqc
- ANSYS Twin Builder Used to Create a Digital Twin of GE Offshore Wind Turbine — https://youtu.be/3Y1InF4_pKY
- The Power of Digital Twins for Building Automation Operations — https://youtu.be/JS_SOIoLmGk
- Siemens PLM — The Real Value of the Digital Twin — <https://www.youtube.com/watch?v=gK5sHDfBMP4>
- Minds + Machines: Meet A Digital Twin — <https://youtu.be/2dCz3oL2rTw>
- Digital twin applied to marketing (Negócios SC) — <https://negociossc.com.br/blog/o-que-e-digital-twin-e-como-seu-conceito-e-aplicado-ao-marketing/>
- ISO 23247-1:2021 — Digital twin framework for manufacturing, Part 1: Overview and general principles — <https://www.iso.org/standard/75066.html>
- ISO/FDIS 23247-6 — Digital twin framework for manufacturing, Part 6: Digital twin composition (2026) — <https://committee.iso.org/standard/87426.html>
- Implementation of ISO 23247 for Digital Twins in Production Systems: Towards AI-Based Sustainable and XR-Based Human-Centric Manufacturing (2026) — <https://research.chalmers.se/en/publication/552658>
- Industrial Metaverse Digital Twin: ISO 23247 Compliant Architecture for AI-Driven Simulation — <https://bibliotecadigital.ipb.pt/entities/publication/7f62c145-ed9a-4e6f-86e9-2f71f7531319/full>